Mines Paris PSL - Etilab

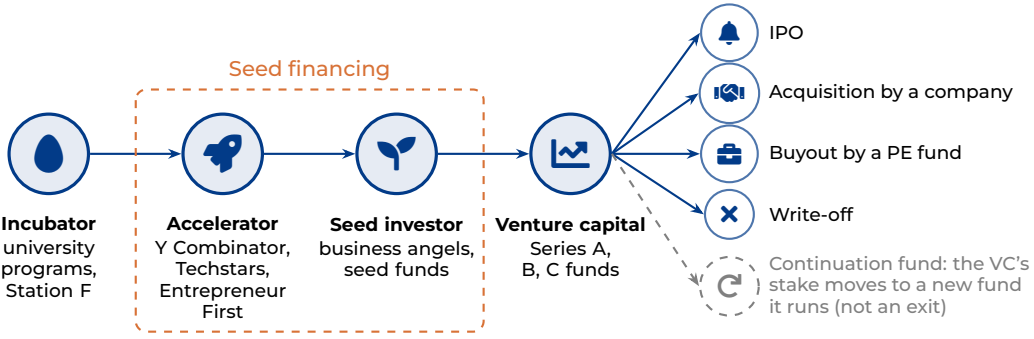
Protection and Dilution: SAFEs and MFN Clauses in Seed Financing

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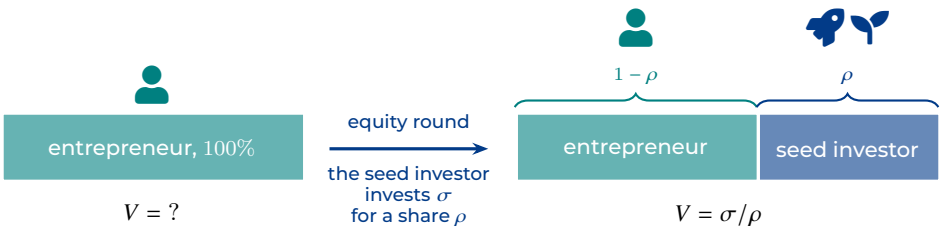
The life of a venture-backed startup

A startup raises money from a sequence of investors before it exits.



Exit routes follow Cumming and MacIntosh (2003): IPO, acquisition, secondary sale, buyback, write-off.

A first equity round combines the first issuance of N shares at price p.



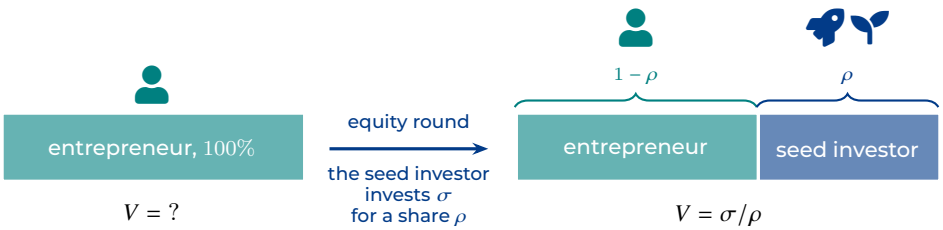
Definition

An equity round is characterized by (σ, ρ, N, p) such that $Np = \sigma/\rho = V$.

A seed investor invests \$120K for 7% equity, so $V = \$120\text{K}/0.07 \approx \$1.71\text{M}$.

Contracts in seed financing
The equity round

A first equity round combines the first issuance of N shares at price p.



Definition

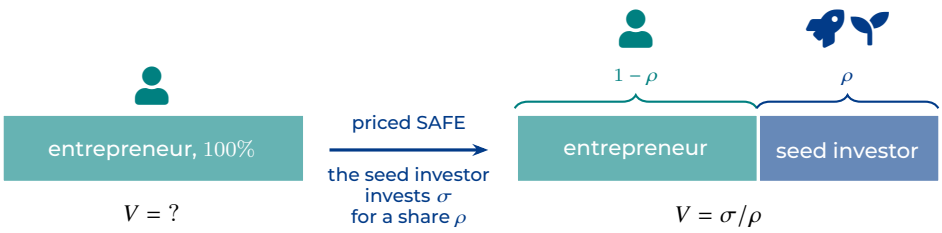
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But equity rounds are administratively heavy (new share class, legal fees).

Contracts in seed financing
Priced SAFE (Simple Agreement for Future Equity)

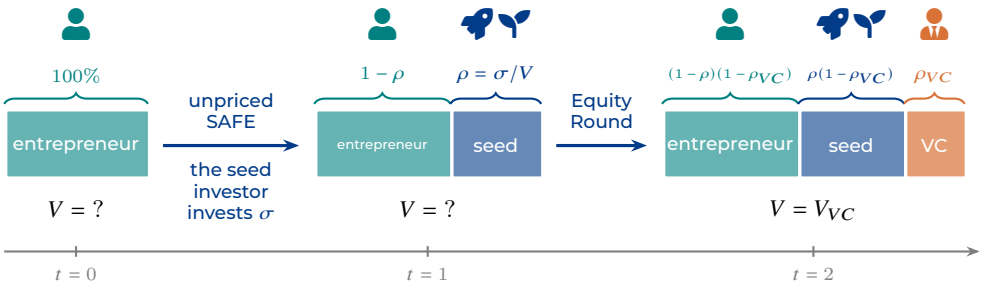
A first priced SAFE defines the project's first valuation without issuance of shares.



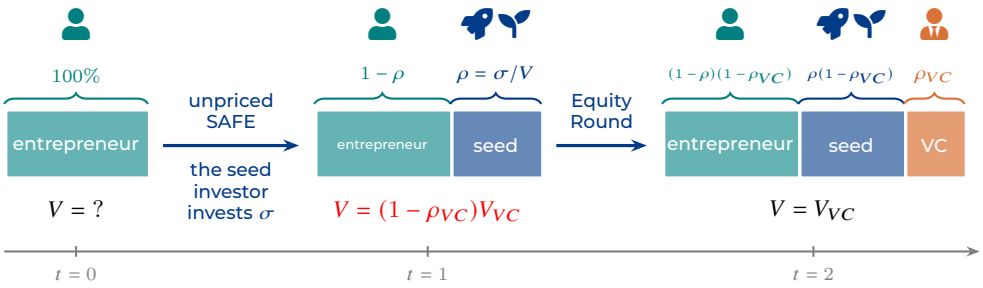
A seed investor invests \$125K for 7% on a SAFE, so $V = \$125\text{K}/0.07 \approx \$1.79\text{M}$.

Contracts in seed financing
Unpriced SAFE with no cap and no discount

A first unpriced SAFE with no cap and no discount leaves the valuation to the next round.

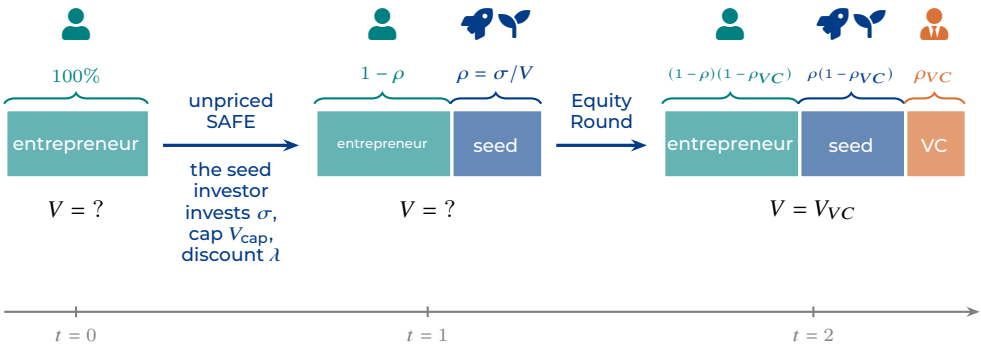


A first unpriced SAFE with no cap and no discount leaves the valuation to the next round.



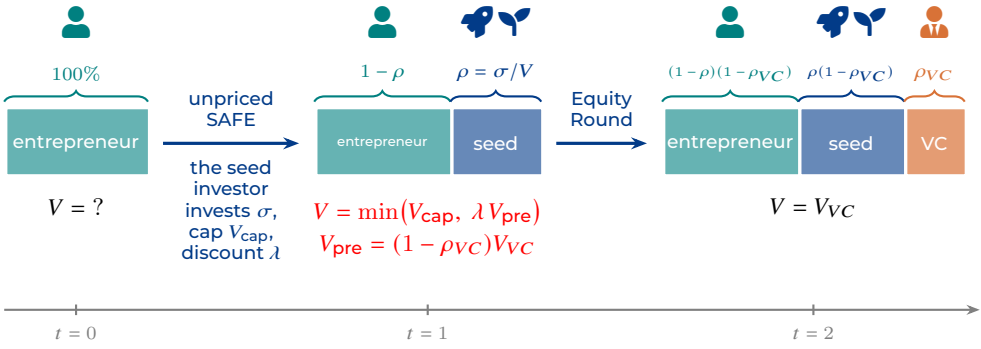
Contracts in seed financing
Unpriced SAFE with a cap V_{cap} and a discount λ

A first unpriced SAFE defines a valuation cap and a discount without issuance of shares.



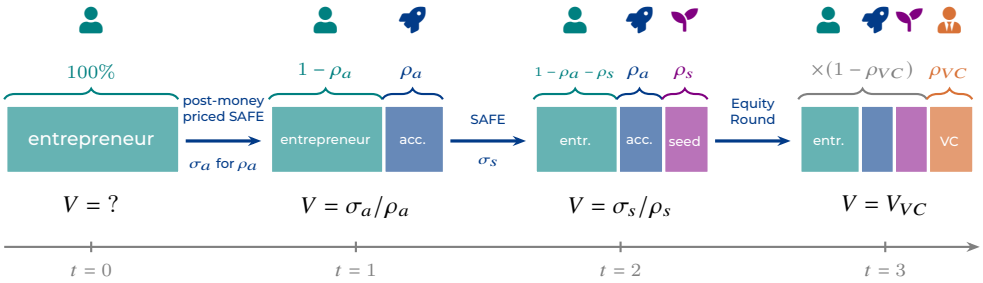
Contracts in seed financing
Unpriced SAFE with a cap V_{cap} and a discount λ

A first unpriced SAFE defines a valuation cap and a discount without issuance of shares.



Contracts in seed financing
The accelerator's post-money priced SAFE

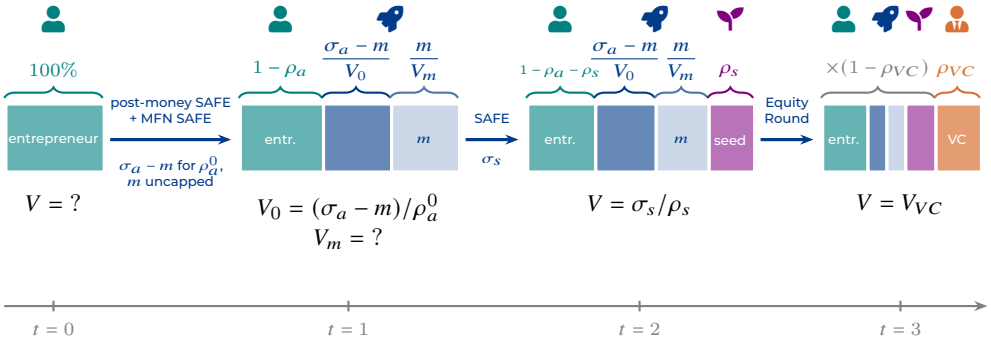
The accelerator's post-money priced SAFE fixes its share ρ_a until the equity round.



"Post-money" ensures that other SAFEs cannot dilute the accelerator's SAFE.

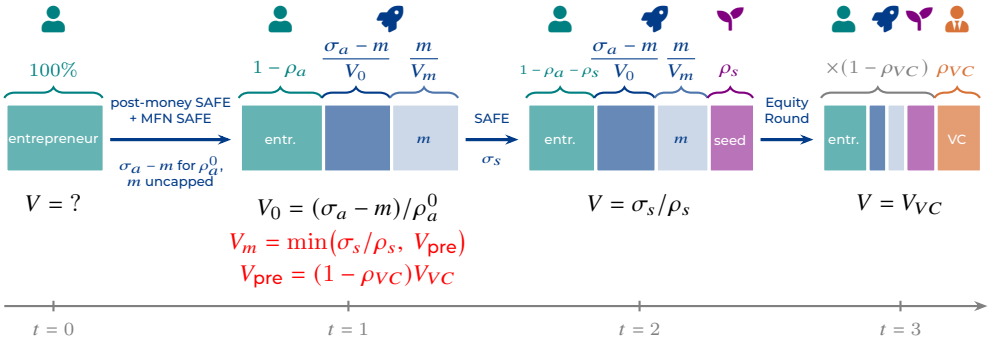
Contracts in seed financing
Post-money SAFE and MFN SAFE

The accelerator can add to its post-money SAFE an MFN SAFE, which converts on the best upcoming terms.



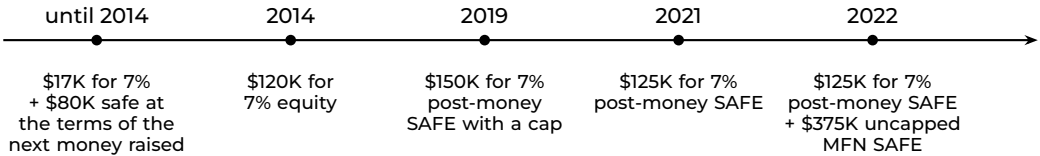
The accelerator's SAFEs convert at two valuations. The MFN SAFE buys m/V_m: it takes more from a seed round with a low valuation.

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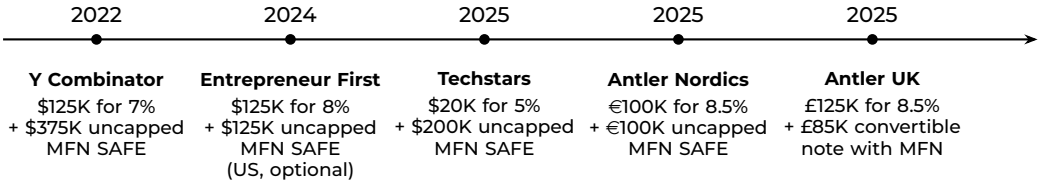
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Chronology of Y Combinator deal terms



Accelerators' MFN SAFEs

Y Combinator launched a trend among accelerators.



A convertible note works like a SAFE, but it is debt: it carries an interest rate and a maturity, and ranks senior in liquidation.

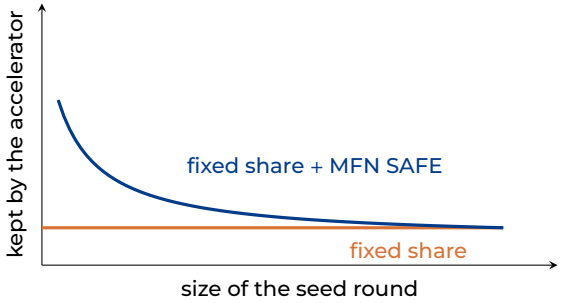
The question

Research question

Is it worth adding an MFN SAFE to a post-money priced SAFE?

- For the accelerator: does the clause raise its profit, and how should it split its money between the two SAFEs?
- For the projects: does it raise the probability of a seed round? Of an equity round?
- Across projects: how do these anti-dilution clauses separate software and deep-tech projects?

The idea: an MFN SAFE takes more from small seed rounds



- The MFN SAFE converts at the seed investor's price. A small round sets a low price, so the same amount buys a larger share.
- A fixed share takes the same from every round: to protect more, it must give up the largest rounds.

Findings

Any fixed share with a contribution is beaten by a small MFN component.

For the accelerator

- It takes more from small rounds.
- The seed investor overinvests on them.
- All the money goes through the clause.

For the projects

- A protection shortens the seed zone.
- The contribution moves it either way.
- Overinvestment raises full funding.

Software and deep tech

- A seed round covers at most $W(I) < I$.
- Software: the clause is best.
- Beyond: a share for almost no money, a quarter of the surplus lost.

Related literature

The agents

The project

- A liquidity shock $\ell \sim g$.
- A scaling cost I.
- A value $\pi \sim f$.

(A1) $E(\pi) < I$.

(A2) f, g positive, log-concave.

The accelerator

- Owns the whole project.
- Spends σ_a before ℓ and π.
- Can add anti-dilution clauses.

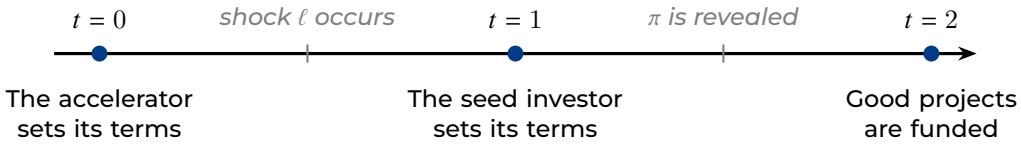
The seed investor

- Invests σ_s after ℓ, before π.
- Makes TIOLI offers.
- Runs an equity round, or a priced or unpriced SAFE.

Experimentation

Covering the liquidity shock ℓ finances an experiment, which reveals the project's value π. The accelerator and the seed investor are the investors able to finance it.

Timing



- The accelerator's contribution is spent before the shock: if it rejects the seed round at $t = 1$, the project is abandoned and it gets nothing.
- If the project is not funded at $t = 2$, it returns nothing, and all the money invested is lost.

We solve the game by backward induction.

Throughout

These hold in every result that follows.

- (A1) $E(\pi) < I$ and (A2) f, g positive and log-concave.
- The seed investor enters only when the shock exceeds the contribution, $\ell > \sigma_a$, and the remaining need stays in $[0, 1]$, $x \in [0, 1]$.
- Every SAFE converts at full funding, at the pre-money value $\pi - x$.
- The accelerator's protection comes first. As in YC's post-money SAFE, it is not diluted by the SAFEs signed after it, and the later SAFEs share what it leaves.

Benchmark: no clause, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds no protection. Seed holds a priced SAFE.

Suppose $\ell > \sigma_a$ and $x \in [0, 1]$. The seed investor solves

$$\begin{aligned} \max_{\sigma_s, \rho_s} \quad & \Pi_s = \rho_s\, W(x) - \sigma_s \\ \text{under} \quad & \sigma_s \geq \ell - \sigma_a \qquad (L) \end{aligned}$$

with $\rho_s \in [0, 1]$.

Benchmark: no clause, priced SAFE

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Accelerator holds no protection. Seed holds a priced SAFE.

- The seed investor exactly covers the remaining liquidity cost, $\sigma_s^* = \ell - \sigma_a$, and $x^* = I$.
- It takes all of the equity, $\rho_s^* = 1$.
- It earns $\Pi_s^* = W(I) - (\ell - \sigma_a)$, and finances $\ell \leq \sigma_a + W(I)$.
- The accelerator keeps nothing.

Benchmark: no clause, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds no protection. Seed holds a priced SAFE.

The accelerator's contribution is spent before the shock. It solves

$$\max_{\sigma_a \geq 0} \quad \Pi_a = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell - \sigma_a.$$

Benchmark: no clause, priced SAFE

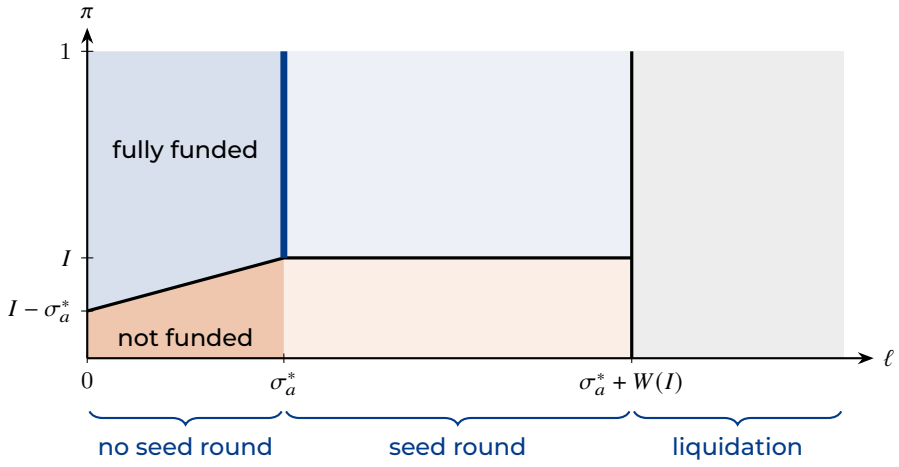
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Accelerator holds no protection. Seed holds a priced SAFE.

Every optimal contribution σ_a^* is zero or solves

$$W(I)\, g(\sigma_a^*) = \big(1 - G(\sigma_a^*)\big) + \int_0^{\sigma_a^*} F(I + \ell - \sigma_a^*)\, g(\ell)\, d\ell.$$

Benchmark: no clause, priced SAFE



Uncapped SAFE | With discount | Capped SAFE

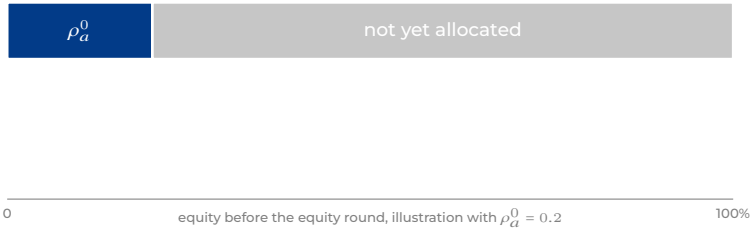
The seed investor's answer to each protection

Accelerator	Last shock the seed investor finances	Accelerator keeps on each round
No clause	$\sigma_a + W(I)$	nothing
Fixed share ρ_a	$\sigma_a + (1-\rho_a)W(I)$	$\rho_a W(I)$
Own SAFE (V_a, λ_a)	$\sigma_a + W\big(I + \frac{\sigma_a}{\lambda_a}\big) - \frac{\sigma_a}{V_a} W\big(I + \frac{V_a}{\lambda_a}\big)$	$W(I) - W\big(I + \frac{\sigma_a}{\lambda_a}\big) + \frac{\sigma_a}{V_a} W\big(I + \frac{V_a}{\lambda_a}\big)$
MFN clause (ρ_a^0, m)	$\sigma_a + (1-\rho_a^0)W(I) - m$	at least $\rho_a^0 W(I)$, and $\rho_a^0 W(I) + m$ on the last round

Without the MFN clause, the seed investor invests its need, and the accelerator keeps the same value on every round.

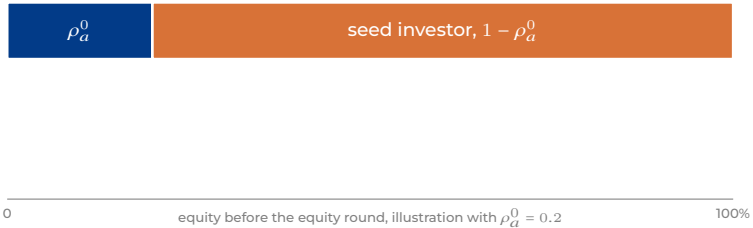
▸ Fixed share, priced ▸ Fixed share, capped ▸ Own SAFE ▸ MFN clause

How the MFN clause splits the equity



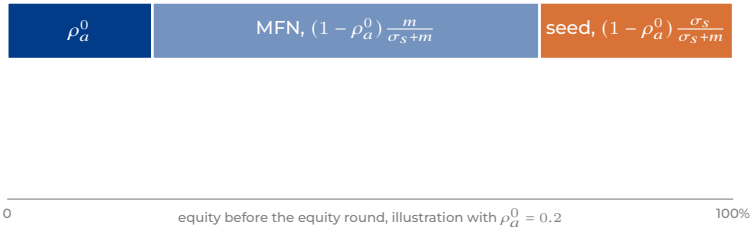
1. The accelerator secures its fixed part ρ_a^0.

How the MFN clause splits the equity



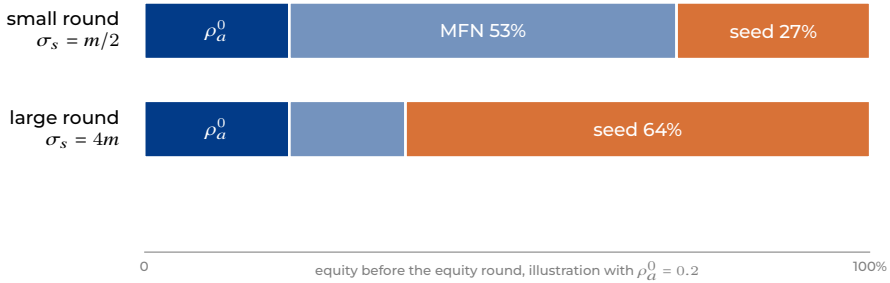
1. The accelerator secures its fixed part ρ_a^0.
2. Without the clause, the seed investor takes all the rest.

How the MFN clause splits the equity



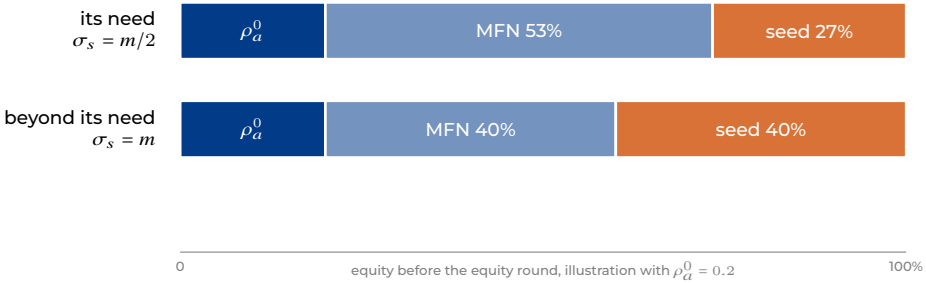
1. The accelerator secures its fixed part ρ_a^0.
2. Without the clause, the seed investor takes all the rest.
3. With the clause, the rest is shared in proportion to σ_s and m.

How the MFN clause splits the equity



1. The accelerator secures its fixed part ρ_a^0.
2. Without the clause, the seed investor takes all the rest.
3. With the clause, the rest is shared in proportion to σ_s and m.
4. On a small round, the seed investor takes little.

How the MFN clause splits the equity



1. The accelerator secures its fixed part ρ_a^0.
2. Without the clause, the seed investor takes all the rest.
3. With the clause, the rest is shared in proportion to σ_s and m.
4. On a small round, the seed investor takes little.
5. It answers by investing beyond its need.

The seed investor prefers an unpriced SAFE

With an unpriced SAFE, the seed investor and the MFN amount convert at the same price in every state.

Against the MFN clause, the seed investor's share is capped at

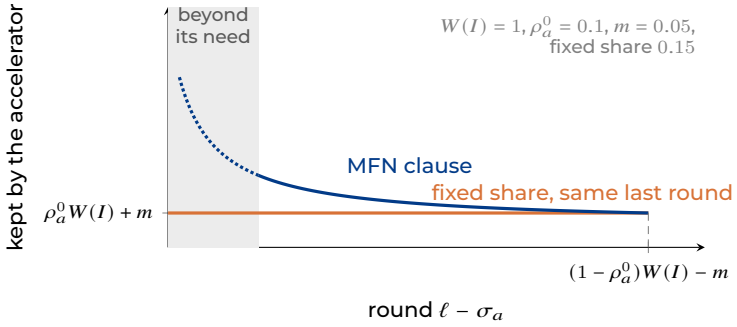
$$\rho_s = \frac{(1-\rho_a^0)\,\sigma_s}{\sigma_s + m},$$

reached in every state by the cap $V_s = (\sigma_s + m)/(1-\rho_a^0)$.

With a priced SAFE, the MFN amount converts at a lower price in low states.

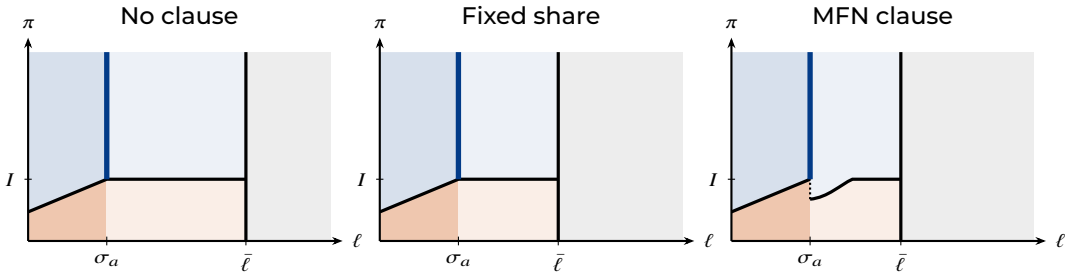
▸ Derivation ▸ Priced SAFE

The MFN clause keeps more on every round but the last



With the same last round, the MFN clause keeps $\rho_a^0 W(I) + (1 - \rho_a^0)\, m\, W(I)/(\ell - \sigma_a + m)$. On the smallest rounds, the seed investor invests beyond its need to dilute the MFN amount.

The MFN clause bends the seed round



Above the black line, the project is fully funded. Under the MFN clause, the seed investor's extra money lowers the threshold below I on the smallest rounds.

The accelerator's problem under each protection

The accelerator chooses its contribution $\sigma_a \geq 0$ and its protection to maximize

$$\Pi_a = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell - \sigma_a + \text{what it keeps on the seed rounds.}$$

Accelerator	What it keeps on the seed rounds	Constraints
No clause	0	
Fixed share	$\rho_a W(I)\big[G\big(\sigma_a + (1-\rho_a)W(I)\big) - G(\sigma_a)\big]$	$\rho_a \in [0,1]$
Own SAFE	the same, with $\rho_a W(I)$ replaced by the value in the previous table	$V_a \geq \sigma_a$, $\lambda_a \in (0,1]$
MFN clause	$\int_{\sigma_a}^{\sigma_a + (1-\rho_a^0)W(I) - m} \big[W(I) - (\ell - \sigma_a) - \Pi_s^*(\ell) - \int_{x^*(\ell)}^{I} F\big]\, g(\ell)\, d\ell$	$\rho_a^0 \in [0,1]$, $0 \leq m \leq \sigma_a$

A small MFN component beats any fixed share

Take any fixed share with $\sigma_a > 0$ and $0 < \rho_a < 1$. Adding a small MFN amount m strictly raises the accelerator's profit.

On round $\ell - \sigma_a$, the value the accelerator keeps rises by

$$\frac{(1 - \rho_a^0)\, m\, W(I)}{\ell - \sigma_a + m}.$$

Divided by m, this gain is unbounded near the smallest rounds, and the loss stays bounded.

Not every MFN clause does better: a large m shortens the zone.

▸ Proof

A capped SAFE does no better than a fixed share

Against the accelerator's own capped SAFE, the seed investor invests its need. Only the value kept at the threshold I matters, and a fixed share can keep the same value.

The SAFE keeps at least $W(I) - W(I + \sigma_a)$. It matches the fixed share when the optimal fixed share keeps more, as in our simulations.

▸ Derivation ▸ Own SAFE in the clause

All of the accelerator's money goes through the clause

When $m^* = \sigma_a^*$, moving protection from the fixed part into the clause still pays. The constraint $m \leq \sigma_a$ stops it.

- The fixed part curbs the seed investor's investment beyond its need.
- In our simulations, $m^* = \sigma_a^*$ in all cases but one.
- Without a contribution, the clause is a fixed share, and the fixed share may do better.

Money beyond the need is lost

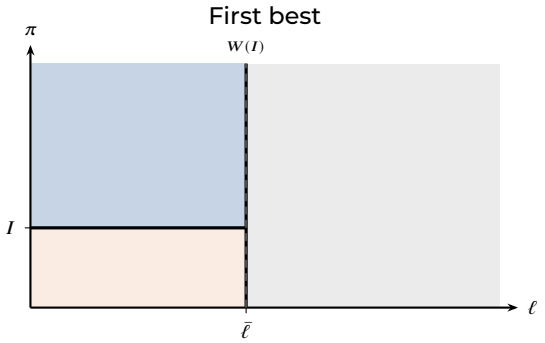
A covered project with threshold $x \leq I$ has surplus

$$W(x) - (I + \ell - x) \;=\; W(I) - \ell \;-\; \underbrace{(I - x)\, F(x)}_{\text{lost when } \pi < x} \;-\; \underbrace{\int_x^I (I - \pi)\, f(\pi)\, d\pi}_{\text{funds projects with } \pi < I} \;.$$

An abandoned project has surplus $-\sigma_a$: the contribution is spent before the shock.

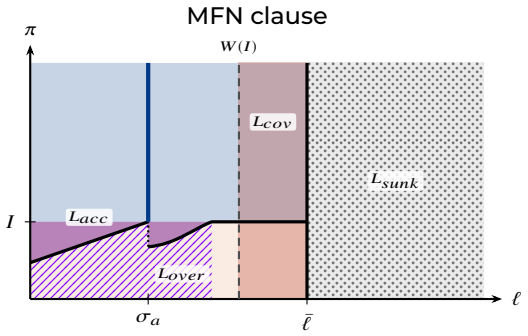
Every dollar brought beyond the need, $I - x$, is either lost or funds a project worth less than its scaling cost.

The first best brings no money before the shock



It covers exactly the shocks $\ell \le W(I)$, at the threshold I: $TS^* = \int_0^{W(I)} \big(W(I) - \ell\big) g(\ell)\, d\ell$.
Without protection and without contribution, the seed investor reaches it.

Four losses from the first best

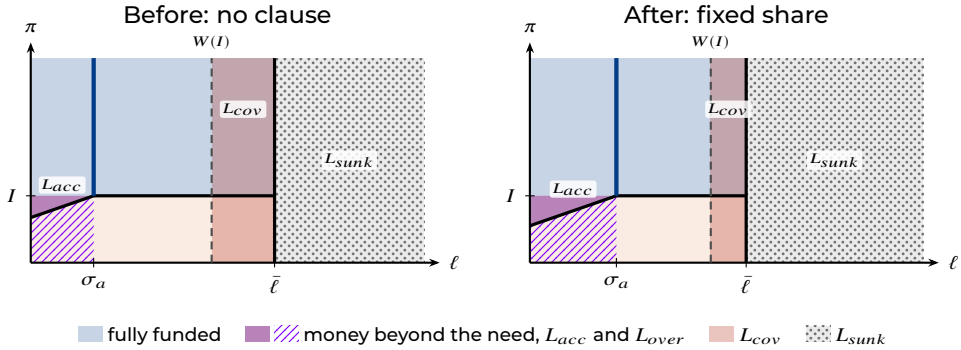


fully funded | money beyond the need, L_{acc} and L_{over} | L_{cov} | L_{sunk}

$$TS = TS^* - L_{acc} - L_{over} - L_{cov} - L_{sunk}$$

▸ Derivation

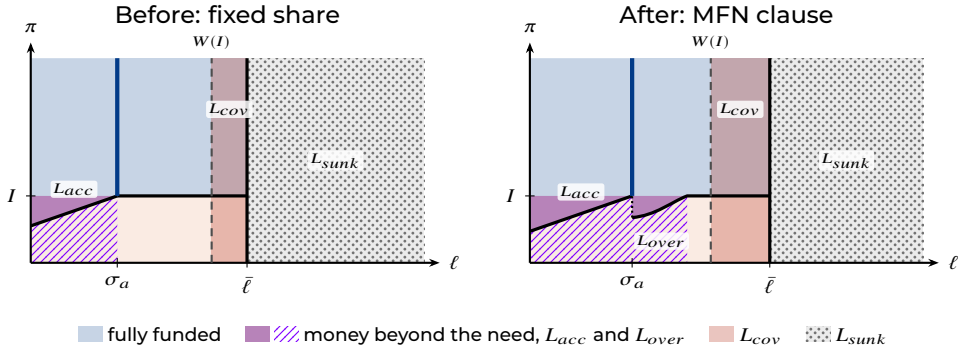
Total surplus, before and after a fixed share



The fixed share abandons the last shocks. Covering them was worth $\sigma_a + W(I) - \ell \geq 0$ more than abandoning, since the contribution is spent. More contribution lowers the total surplus.

▸ Derivation

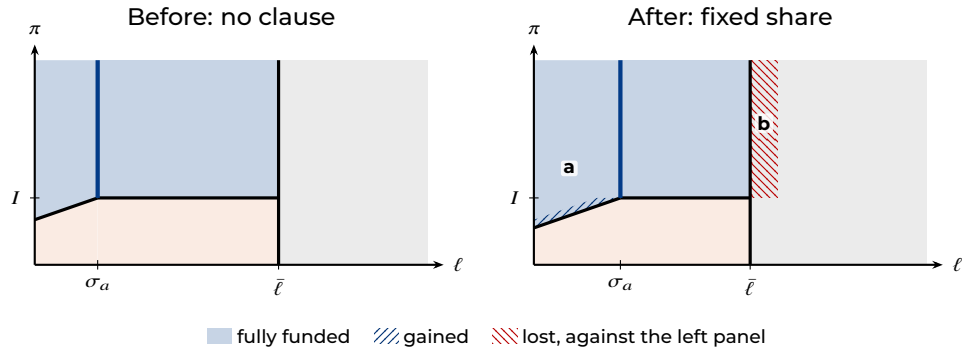
Total surplus, before and after the MFN clause



The MFN clause adds L_{over} on the smallest rounds. The last project and the contribution move the other losses either way.

▸ Derivation

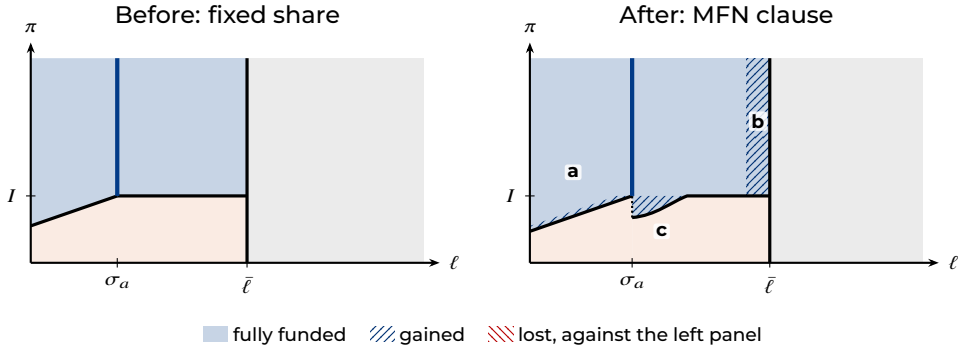
Full funding, before and after a fixed share



(a) More money lowers the threshold on the shocks the accelerator covers alone. (b) The seed investor stops earlier. If the fixed share does not raise the contribution, (a) is a loss.

▸ Derivation

Full funding, before and after the MFN clause



(a) The contribution moves either way, drawn higher here. (b) The last project moves either way, drawn as a gain. (c) The money beyond the need always raises full funding.

▸ Derivation

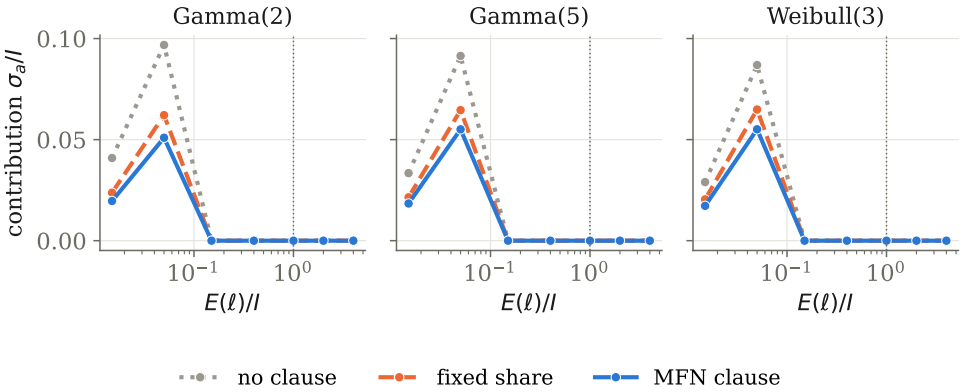
Each protection moves the outcomes either way

	Term	Fixed share against no clause	MFN clause against fixed share
Full funding	contribution	sign of $\sigma_a^P - \sigma_a^B$	sign of $\sigma_a^M - \sigma_a^P$
	last project	< 0	sign of $\rho_a^P W(I) - \rho_a^{0M} W(I) - m^M$
	beyond the need	none	≥ 0
Seed round	contribution	no sign	no sign
	last project	< 0	sign of $\rho_a^P W(I) - \rho_a^{0M} W(I) - m^M$
Total surplus	contribution	sign of $\sigma_a^B - \sigma_a^P$	no sign
	last project	< 0	sign of $\rho_a^P W(I) - \rho_a^{0M} W(I) - m^M$
	beyond the need	none	≤ 0

The contribution moves every outcome, in a direction the theory does not sign.

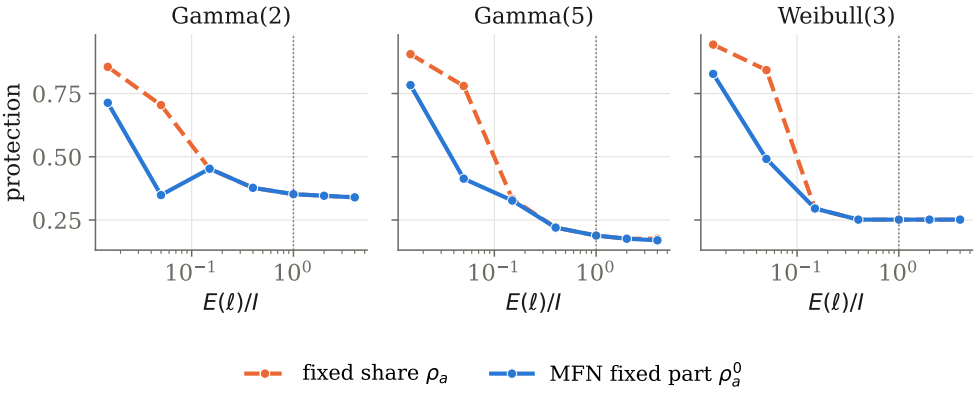
▸ Full funding ▸ Fixed share ▸ MFN clause ▸ Seed round bound ▸ MFN, seed rounds

The accelerator invests only when liquidity needs are small



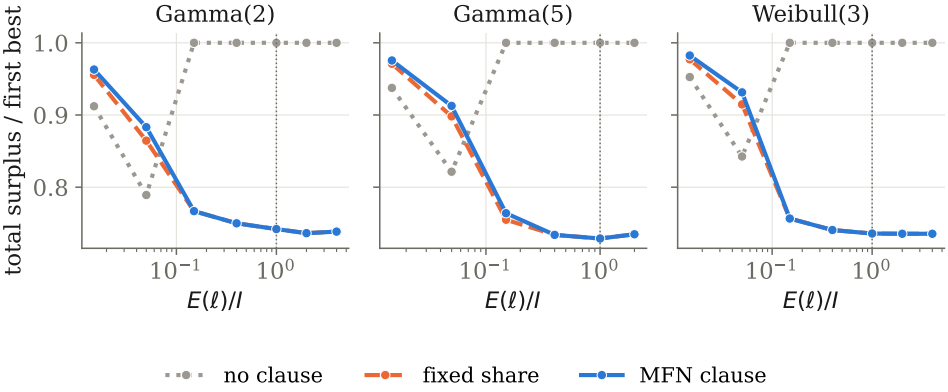
Software, $E(\ell)/I \leq 0.05$: the accelerator invests, less with a protection, and almost all of its money goes through the clause. From $E(\ell)/I = 0.15$ on, its contribution vanishes under any protection.

Without a contribution, the protection is a fixed share



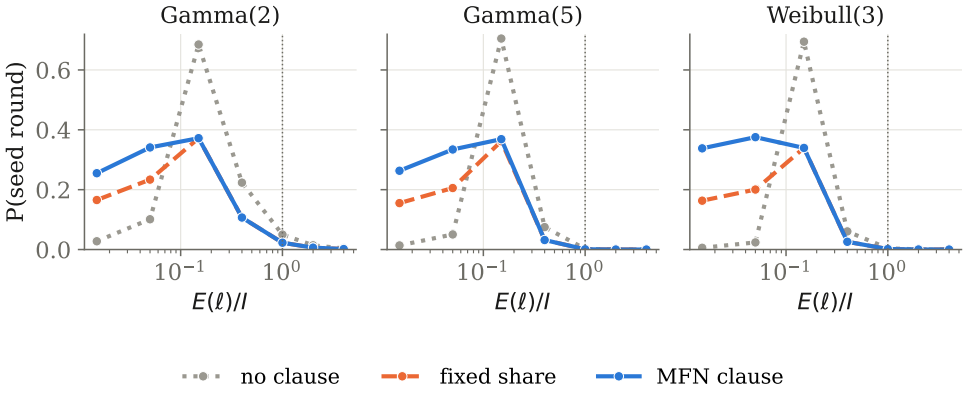
As σ_a tends to zero, so does the MFN amount, $m \leq \sigma_a$. The accelerator buys a fixed share with an amount and at a valuation that tend to zero.

Welfare: total surplus



Software: MFN clause > fixed share > no clause. Beyond: no clause reaches the first best, and the protection loses about a quarter of it.

Which projects survive: seed rounds



Software: the protection lowers the contribution and leaves more shocks to the seed investor. Beyond: it only shortens the seed investor's zone.

▸ Full funding

Takeaways

Main result

Any fixed share with a contribution is beaten by a small MFN component, which takes more from small rounds.

- The seed investor answers with an unpriced SAFE and invests beyond its need.
- In software, all of the money goes through the clause.
- Beyond software, the accelerator buys its share for almost no money, and this share costs a quarter of the surplus.

▸ Robustness ▸ Extensions

Thank you

Protection and Dilution:
SAFEs and MFN Clauses in Seed Financing

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Robustness: another distribution of π

With a Beta$(2, 2)$ value on $[0, 1.8]$ instead of a uniform, nothing changes.

- Software: MFN clause $>$ fixed share $>$ no clause, all the money through the clause, total surplus 0.82–0.97 of the first best with the clause.
- Deep tech: the contribution vanishes, the fixed share is 0.18 to 0.42, and the total surplus falls to 0.74–0.76 of the first best.

Back

Robustness: the founders keep a stake

The founders keep ρ_e before the seed round.

- The seed investor's answers hold with $\rho_e + \rho_a$ instead of ρ_a.
- Without protection, the first best is no longer reached.
- With $\rho_e = 0.5$, the optimal fixed share comes with a contribution that tends to zero, while the MFN clause keeps the accelerator investing: total surplus 0.79–0.87 of the first best, against 0.62–0.67.

Back

Extension: a priced SAFE against the MFN clause

If the accelerator could require a priced SAFE, a small MFN component would still beat any fixed share with a contribution.

The seed investor's share never exceeds $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$, whatever its SAFE, and the seed investor stops financing only rounds near the ends of its zone.

Proof Back

Extension: the accelerator's own SAFE as the fixed part

With its own SAFE as the fixed part, the optimal clause earns at most $2 \cdot 10^{-5}$ more than with a fixed share.

- The seed investor invests its need on almost every round, so only the value at the threshold I matters.
- Y Combinator used both forms of its 7%: a capped post-money SAFE from 2019, without MFN, and today a fixed 7% with the MFN SAFE.

Derivation Back

Related literature

Literature	Key papers	What we know	What this paper adds
Experimentation	Kerr, Nanda, Rhodes-Kropf (2014) Ewens, Nanda, Rhodes-Kropf (2018) Bergemann, Hege (1998, 2005)	Value is learned by investing, so early money funds an experiment.	The terms of the second investor, once the first has invested.
VC securities	Admati, Pfleiderer (1994) Kaplan, Strömberg (2003) Coyle, Green (2014)	Which security aligns investors. SAFEs replaced notes at the seed stage.	Priced against unpriced SAFEs, after an accelerator.
MFN clauses	Cooper (1986) Besanko, Lyon (1993) McAfee, Schwartz (1994)	A buyer holds the clause against a seller, and it often fails to protect.	An investor holds it against a later investor, and it works as price discrimination.
Accelerators	Cohen, Hochberg (2014) Gonzalez-Uribe, Leatherbee (2017) Hallen, Cohen, Bingham (2020)	What accelerators do for startups.	The accelerator's contract.

To our knowledge, the first model of an accelerator's protection against seed dilution.

Back

No clause, uncapped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds no protection. Seed holds an uncapped SAFE without discount.

Suppose $\ell > \sigma_a$ and $x \in [0, 1]$. The seed investor solves

$$\begin{aligned} \max_{\sigma_s} \quad & \Pi_s = \int_x^1 \rho_s(\pi)\, (\pi - x)\, f(\pi)\, d\pi - \sigma_s \\ \text{under} \quad & \sigma_s \geq \ell - \sigma_a \end{aligned} \tag{L}$$

with $\rho_s(\pi) = \min\{\sigma_s/(\pi - x),\ 1\}$: the SAFE pays $\min(\sigma_s, \pi - x)$ in state π.

No clause, uncapped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds no protection. Seed holds an uncapped SAFE without discount.

- Since $\min(\sigma_s, \pi - x) = (\pi - x) - (\pi - x - \sigma_s)^+$, the SAFE pays $W(x) - W(x + \sigma_s)$ in expectation, and at most its amount in every state.
- The seed investor loses money at every amount, $\Pi_s = -\int_x^{x+\sigma_s} F(t)\, dt < 0$.
- No shock $\ell > \sigma_a$ obtains a seed round, and the accelerator keeps nothing on these projects.

No clause, uncapped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds no protection. Seed holds an uncapped SAFE without discount.

The accelerator's contribution is spent before the shock. It solves

$$\max_{\sigma_a \geq 0} \quad \Pi_a = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell - \sigma_a.$$

No clause, uncapped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

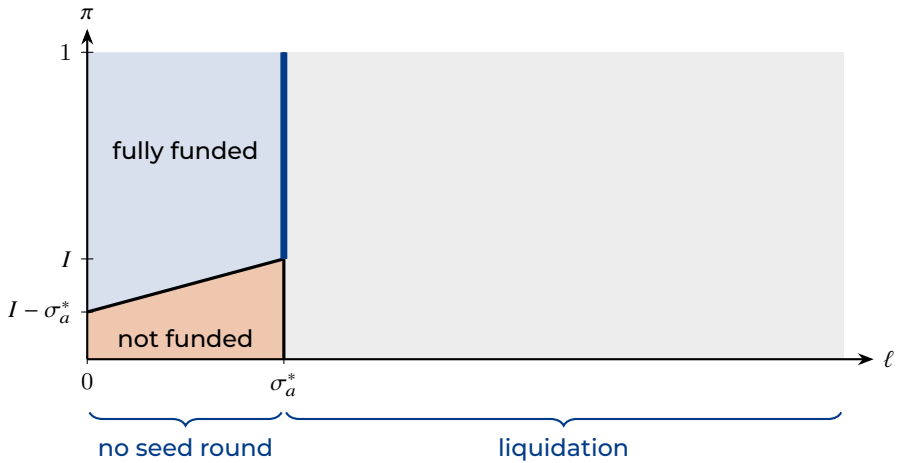
Accelerator holds no protection. Seed holds an uncapped SAFE without discount.

Every optimal contribution σ_a^* is zero or solves

$$W(I)\, g(\sigma_a^*) = \big(1 - G(\sigma_a^*)\big) + \int_0^{\sigma_a^*} F(I + \ell - \sigma_a^*)\, g(\ell)\, d\ell.$$

It is the benchmark contribution.

No clause, uncapped SAFE



Back

No clause, uncapped SAFE with discount

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds no protection. Seed holds an uncapped SAFE with discount λ.

Suppose $\ell > \sigma_a$ and $x \in [0, 1]$. The seed investor solves

$$\begin{aligned} \max_{\sigma_s, \lambda} \quad & \Pi_s = \int_x^1 \rho_s(\pi)\, (\pi - x)\, f(\pi)\, d\pi - \sigma_s \\ \text{under} \quad & \sigma_s \geq \ell - \sigma_a \end{aligned} \tag{L}$$

with $\rho_s(\pi) = \min\{\sigma_s/(\lambda(\pi - x)),\ 1\}$ and $\lambda \in (0, 1]$: the SAFE pays $\min(\sigma_s/\lambda, \pi - x)$ in state π.

No clause, uncapped SAFE with discount

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds no protection. Seed holds an uncapped SAFE with discount λ.

- Since $\min(\sigma_s/\lambda, \pi - x) = (\pi - x) - (\pi - x - \sigma_s/\lambda)^+$, the SAFE pays $W(x) - W(x + \sigma_s/\lambda)$ in expectation.
- The seed investor exactly covers the remaining liquidity cost, $\sigma_s^* = \ell - \sigma_a$, and $x^* = I$.
- The discount $\lambda^* = \min\{1, (\ell - \sigma_a)/(1 - I)\}$ gives it all of the equity.
- It earns $\Pi_s^* = W(I) - (\ell - \sigma_a)$, and finances $\ell \le \sigma_a + W(I)$.

No clause, uncapped SAFE with discount

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds no protection. Seed holds an uncapped SAFE with discount λ.

The accelerator's contribution is spent before the shock. It solves

$$\max_{\sigma_a \geq 0} \quad \Pi_a = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell - \sigma_a.$$

No clause, uncapped SAFE with discount

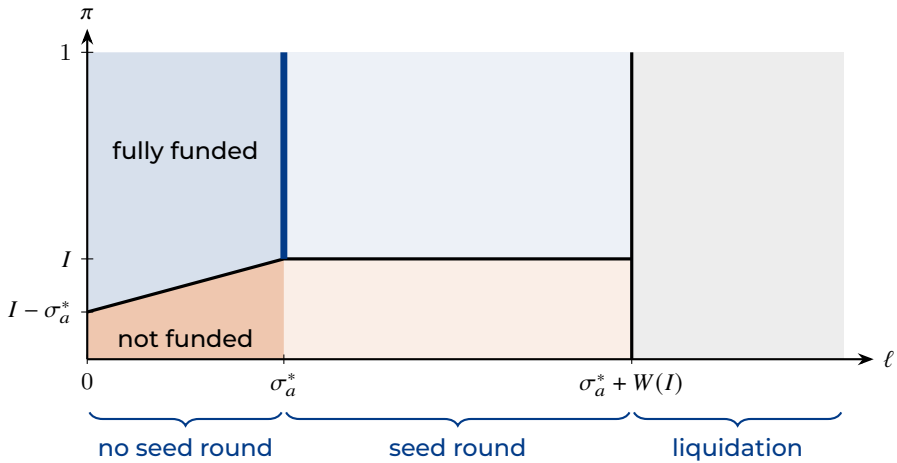
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Accelerator holds no protection. Seed holds an uncapped SAFE with discount λ.

Every optimal contribution σ_a^* is zero or solves

$$W(I)\, g(\sigma_a^*) = \big(1 - G(\sigma_a^*)\big) + \int_0^{\sigma_a^*} F(I + \ell - \sigma_a^*)\, g(\ell)\, d\ell.$$

No clause, uncapped SAFE with discount



Back

No clause, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds no protection. Seed holds a capped SAFE with discount.

Suppose $\ell > \sigma_a$ and $x \in [0, 1]$. The seed investor solves

$$\max_{\sigma_s, V_s, \lambda} \quad \Pi_s = \int_x^1 \rho_s(\pi)\, (\pi - x)\, f(\pi)\, d\pi - \sigma_s$$
$$\text{under} \quad \sigma_s \geq \ell - \sigma_a \qquad (L)$$

with $\rho_s(\pi) = \min\{\sigma_s / \min(V_s, \lambda(\pi - x)),\ 1\}$ and $\lambda \in (0, 1]$.

No clause, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds no protection. Seed holds a capped SAFE with discount.

- The seed investor exactly covers the remaining liquidity cost, $\sigma_s^* = \ell - \sigma_a$, and $x^* = I$.
- The cap $V_s^* = \ell - \sigma_a$ gives it all of the equity.
- It earns $\Pi_s^* = W(I) - (\ell - \sigma_a)$, and finances $\ell \leq \sigma_a + W(I)$.

No clause, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds no protection. Seed holds a capped SAFE with discount.

The accelerator's contribution is spent before the shock. It solves

$$\max_{\sigma_a \geq 0} \quad \Pi_a = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell - \sigma_a.$$

No clause, capped SAFE

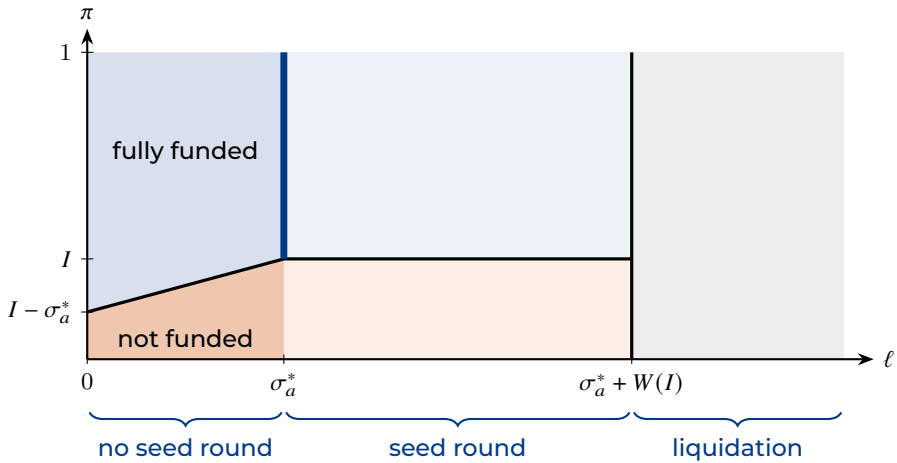
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Accelerator holds no protection. Seed holds a capped SAFE with discount.

Every optimal contribution σ_a^* is zero or solves

$$W(I)\, g(\sigma_a^*) = \big(1 - G(\sigma_a^*)\big) + \int_0^{\sigma_a^*} F(I + \ell - \sigma_a^*)\, g(\ell)\, d\ell.$$

No clause, capped SAFE



Back

Fixed share, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds a fixed share ρ_a. Seed holds a priced SAFE.

Suppose $\ell > \sigma_a$ and $x \in [0, 1]$. The seed investor solves

$$\begin{aligned} \max_{\sigma_s, \rho_s} \quad & \Pi_s = \rho_s\, W(x) - \sigma_s \\ \text{under} \quad & \sigma_s \geq \ell - \sigma_a && (L) \\ & \rho_s \leq 1 - \rho_a && (P) \end{aligned}$$

with $\rho_s \in [0, 1]$.

Fixed share, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds a fixed share ρ_a. Seed holds a priced SAFE.

- The seed investor exactly covers the remaining liquidity cost, $\sigma_s^* = \ell - \sigma_a$, and $x^* = I$.
- The protection binds, $\rho_s^* = 1 - \rho_a$.
- The seed investor earns $\Pi_s^* = (1 - \rho_a)W(I) - (\ell - \sigma_a)$, and finances $\ell \leq \sigma_a + (1 - \rho_a)W(I)$.
- The accelerator keeps $\rho_a W(I)$.

Fixed share, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds a fixed share ρ_a. Seed holds a priced SAFE.

The accelerator solves

$$\max_{\sigma_a, \rho_a} \quad \Pi_a = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell + \int_{\sigma_a}^{\sigma_a + (1-\rho_a)W(I)} \rho_a W(I)\, g(\ell)\, d\ell - \sigma_a.$$

Fixed share, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds a fixed share ρ_a. Seed holds a priced SAFE.

At each σ_a, the optimal share $\rho_a^* \in (0,1)$ is unique and solves

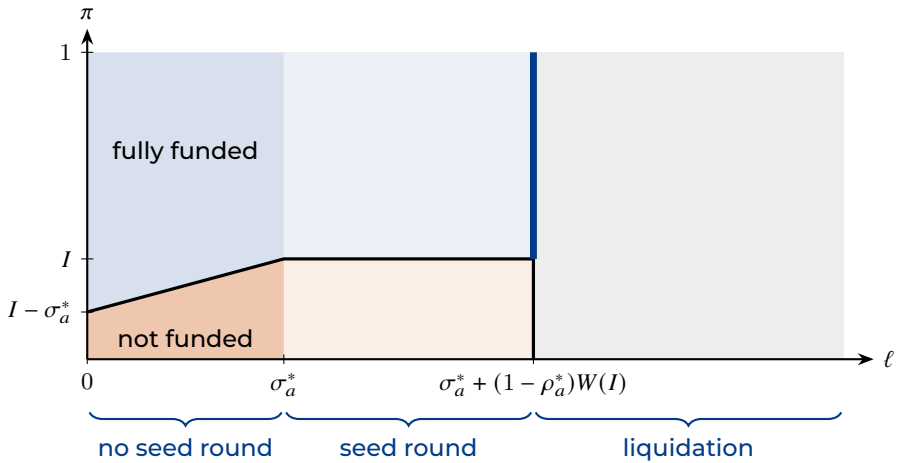
$$G\big(\sigma_a + (1-\rho_a^*)W(I)\big) - G(\sigma_a) = \rho_a^* W(I)\, g\big(\sigma_a + (1-\rho_a^*)W(I)\big).$$

Every optimal contribution σ_a^* is zero or solves

$$(1-\rho_a^*)\, W(I)\, g(\sigma_a^*) + \rho_a^* W(I)\, g\big(\sigma_a^* + (1-\rho_a^*)W(I)\big)$$
$$= \big(1 - G(\sigma_a^*)\big) + \int_0^{\sigma_a^*} F(I + \ell - \sigma_a^*)\, g(\ell)\, d\ell.$$

Compared with the benchmark, the contribution rises if g is higher at the last seed round than at σ_a, and falls otherwise.

Fixed share, priced SAFE



Back

Fixed share, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds a fixed share ρ_a. Seed holds a capped SAFE with discount.

Suppose $\ell > \sigma_a$ and $x \in [0, 1]$. The seed investor solves

$$\begin{aligned} \max_{\sigma_s, V_s, \lambda} \quad & \Pi_s = \int_x^1 \rho_s(\pi)\, (\pi - x)\, f(\pi)\, d\pi - \sigma_s \\ \text{under} \quad & \sigma_s \geq \ell - \sigma_a \end{aligned} \tag{L}$$

with $\rho_s(\pi) = \min\{\sigma_s / \min(V_s, \lambda(\pi - x)),\ \underbrace{1 - \rho_a}_{(P)}\}$ and $\lambda \in (0, 1]$.

Fixed share, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds a fixed share ρ_a. Seed holds a capped SAFE with discount.

- The seed investor obtains exactly what it obtains with a priced SAFE, on the same projects.
- The cap $V_s^* = (\ell - \sigma_a)/(1 - \rho_a)$ gives it the share $1 - \rho_a$ in every state above I.
- The discount plays no role.

Fixed share, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds a fixed share ρ_a. Seed holds a capped SAFE with discount.

The accelerator solves

$$\max_{\sigma_a,\, \rho_a} \quad \Pi_a = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell + \int_{\sigma_a}^{\sigma_a + (1-\rho_a) W(I)} \rho_a W(I)\, g(\ell)\, d\ell - \sigma_a .$$

Fixed share, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
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At each σ_a, the optimal share $\rho_a^* \in (0,1)$ is unique and solves

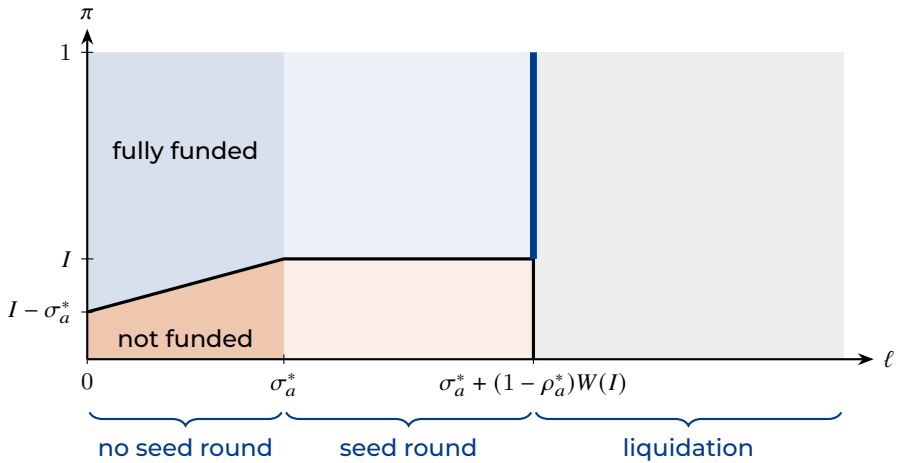
$$G\big(\sigma_a + (1-\rho_a^*)W(I)\big) - G(\sigma_a) = \rho_a^* W(I)\, g\big(\sigma_a + (1-\rho_a^*)W(I)\big).$$

Every optimal contribution σ_a^* is zero or solves

$$(1-\rho_a^*)\, W(I)\, g(\sigma_a^*) + \rho_a^* W(I)\, g\big(\sigma_a^* + (1-\rho_a^*)W(I)\big) = \big(1 - G(\sigma_a^*)\big) + \int_0^{\sigma_a^*} F(I + \ell - \sigma_a^*)\, g(\ell)\, d\ell.$$

Compared with the benchmark, the contribution rises if g is higher at the last seed round than at σ_a, and falls otherwise.

Fixed share, capped SAFE



Back

Accelerator's capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds a capped SAFE on its contribution σ_a, with cap V_a and discount λ_a. Seed holds a priced or capped SAFE.

- YC's $125k for 7% is itself a post-money SAFE.
- The accelerator's SAFE claims $\sigma_a / \min(V_a, \lambda_a(\pi - x))$. Times $\pi - x$, this is the larger of the share σ_a / V_a, when the cap binds, and of the value σ_a / λ_a, when the discount binds.
- The protection comes first, and leaves in expectation

$$W\left(x + \frac{\sigma_a}{\lambda_a}\right) - \frac{\sigma_a}{V_a}\, W\left(x + \frac{V_a}{\lambda_a}\right),$$

since $\int_x^1 (\pi - x - y)^+ f(\pi)\, d\pi = W(x + y)$.

Accelerator's capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds a capped SAFE on its contribution σ_a, with cap V_a and discount λ_a. Seed holds a priced or capped SAFE.

- The seed investor takes all the equity the protection leaves, with a priced SAFE ($\rho_s = 1$) or a capped SAFE ($V_s = \ell - \sigma_a$), and invests only its need.
- It finances the shock if and only if

$$\ell - \sigma_a \le W\Big(I + \frac{\sigma_a}{\lambda_a}\Big) - \frac{\sigma_a}{V_a}\, W\Big(I + \frac{V_a}{\lambda_a}\Big).$$

- The accelerator keeps $W(I)$ minus this value on each of these projects.

Accelerator's capped SAFE

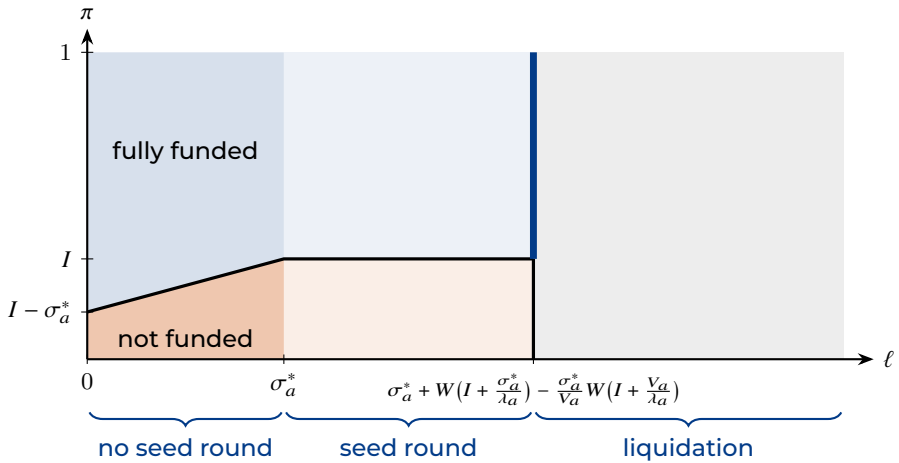
The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds a capped SAFE on its contribution σ_a, with cap V_a and discount λ_a. Seed holds a priced or capped SAFE.

The accelerator's SAFE never does better than the fixed share, and does as well in all our simulations.

- Its profit depends only on what it keeps at the threshold I, as with a fixed share, since the seed investor invests only its need.
- The SAFE keeps at least $W(I) - W(I + \sigma_a)$: it returns its amount when the project allows it. The fixed share has no such minimum.
- When the accelerator invests, the optimal fixed share keeps 0.125 to 0.168, above this minimum, at most 0.028: capping well is then pricing well.

Accelerator's capped SAFE



Back

MFN clause, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds an MFN clause (ρ_a^0, m). *Seed holds a priced SAFE.*

- Each SAFE claims a share of the equity held before the equity round.
- Each claim is multiplied by $\min\{1,\ (1-\rho_a^0)/(\text{sum of the claims})\}$: kept if the claims fit in $1-\rho_a^0$, scaled down to fill it otherwise.
- The seed investor claims ρ_s. The amount m converts at $\min(\sigma_s/\rho_s,\ \pi - x)$ and claims $m/\min(\sigma_s/\rho_s,\ \pi - x)$.
- If $\pi - x \geq \sigma_s/\rho_s$, the claims sum to $\rho_s(\sigma_s + m)/\sigma_s$, and the seed investor keeps

$$\min\left\{\rho_s,\ \rho_s\,\frac{1-\rho_a^0}{\rho_s(\sigma_s+m)/\sigma_s}\right\} = \min\left\{\rho_s,\ \frac{(1-\rho_a^0)\,\sigma_s}{\sigma_s+m}\right\}.$$

- If $\pi - x < \sigma_s/\rho_s$, they sum to $\rho_s + m/(\pi - x)$, and it keeps

$$\min\left\{\rho_s,\ \rho_s\,\frac{1-\rho_a^0}{\rho_s+m/(\pi-x)}\right\} = \min\left\{\rho_s,\ \frac{(1-\rho_a^0)\,\rho_s(\pi-x)}{\rho_s(\pi-x)+m}\right\}.$$

MFN clause, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds an MFN clause (ρ_a^0, m). Seed holds a priced SAFE.

Suppose $\ell > \sigma_a$ and $x \in [0, 1]$. The seed investor solves

$$\begin{aligned} \max_{\sigma_s, \rho_s} \quad & \Pi_s = \int_x^1 \rho_s(\pi)\,(\pi - x)\, f(\pi)\, d\pi - \sigma_s \\ \text{under} \quad & \sigma_s \geq \ell - \sigma_a \qquad (L) \end{aligned}$$

with

$$\rho_s(\pi) = \begin{cases} \min\{\rho_s,\ (1-\rho_a^0)\,\sigma_s/(\sigma_s + m)\} & \text{if } \pi - x \geq \sigma_s/\rho_s, \\ \min\{\rho_s,\ (1-\rho_a^0)\,\rho_s(\pi - x)/(\rho_s(\pi - x) + m)\} & \text{if } \pi - x < \sigma_s/\rho_s, \end{cases}$$

as derived on the previous slide, where the second term of each $\min$ is the protection (MFN).

MFN clause, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds an MFN clause (ρ_a^0, m). Seed holds a priced SAFE.

- The amount m converts at $\min(\sigma_s/\rho_s,\ \pi - x)$.
- The seed investor's share never exceeds $(1 - \rho_a^0)\sigma_s/(\sigma_s + m)$, and, if $m > 0$, is strictly below it when $\pi - x < \sigma_s/\rho_s$.
- The seed investor finances no shock with $\ell - \sigma_a + m > (1 - \rho_a^0)W(I)$.
- If $m > 0$, it earns strictly less than a capped SAFE gives it.

MFN clause, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds an MFN clause (ρ_a^0, m). Seed holds a priced SAFE.

The accelerator solves

$$\max_{\sigma_a,\, \rho_a^0,\, m \le \sigma_a} \Pi_a = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell + \int_{\sigma_a}^{\bar{\ell}} \left[W(I) - (\ell - \sigma_a) - \Pi_s^*(\ell) - \int_{x^*(\ell)}^{I} F(t)\, dt \right] g(\ell)\, d\ell - \sigma_a,$$

with $\bar{\ell}$ the largest shock the seed investor finances.

MFN clause, priced SAFE

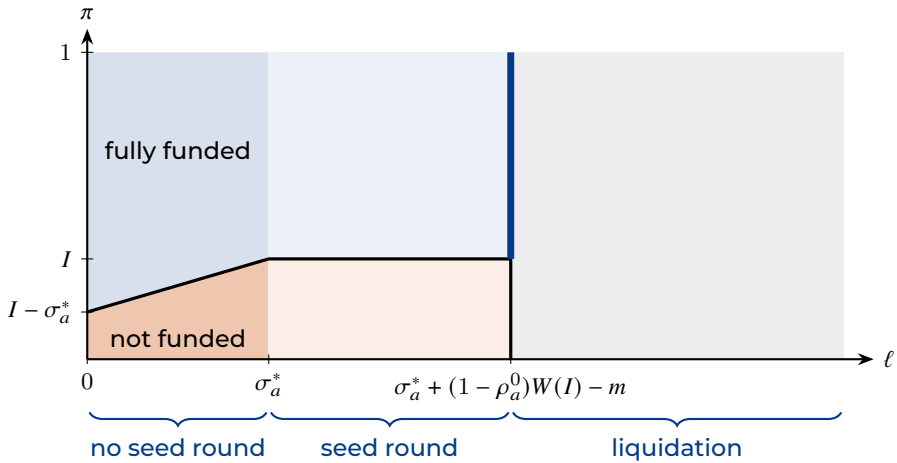
The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds an MFN clause (ρ_a^0, m). Seed holds a priced SAFE.

If the optimal fixed share comes with a contribution, every optimum has $m^* > 0$. If the seed investor finances all the shocks between σ_a^* and $\bar{\ell}$, at given (ρ_a^0, m) with $m < \sigma_a^*$, the contribution solves

$$\left[W(I) - (\bar{\ell} - \sigma_a^*)\right] g(\bar{\ell}) = \left(1 - G(\bar{\ell})\right) + \int_0^{\sigma_a^*} F(I + \ell - \sigma_a^*)\, g(\ell)\, d\ell + \int_{\sigma_a^*}^{\bar{\ell}} \left(\Pi_s^*(\ell) + \int_{x^*(\ell)}^{I} F(t)\, dt \right) g'(\ell)\, d\ell .$$

MFN clause, priced SAFE



Back

MFN clause, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds an MFN clause (ρ_a^0, m). Seed holds a capped SAFE with discount.

- Each SAFE claims a share of the equity held before the equity round.
- Each claim is multiplied by $\min\{1,\ (1-\rho_a^0)/(\text{sum of the claims})\}$: kept if the claims fit in $1-\rho_a^0$, scaled down to fill it otherwise.
- Both SAFEs convert at $\min(V_s, \lambda(\pi - x))$: the seed investor claims $\sigma_s/\min(V_s, \lambda(\pi - x))$, and the claims sum to $(\sigma_s + m)/\min(V_s, \lambda(\pi - x))$.
- The seed investor keeps

$$\min\left\{\frac{\sigma_s}{\min(V_s, \lambda(\pi - x))},\ \frac{\sigma_s}{\min(V_s, \lambda(\pi - x))} \cdot \frac{(1-\rho_a^0)\min(V_s, \lambda(\pi - x))}{\sigma_s + m}\right\},$$

  that is $\min\{\sigma_s/\min(V_s, \lambda(\pi - x)),\ (1-\rho_a^0)\,\sigma_s/(\sigma_s + m)\}$.

MFN clause, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds an MFN clause (ρ_a^0, m). Seed holds a capped SAFE with discount.

Suppose $\ell > \sigma_a$ and $x \in [0, 1]$. The seed investor solves

$$\max_{\sigma_s, V_s, \lambda} \quad \Pi_s = \int_x^1 \rho_s(\pi)\,(\pi - x)\, f(\pi)\, d\pi - \sigma_s$$
$$\text{under} \quad \sigma_s \geq \ell - \sigma_a \qquad (L)$$

with $\rho_s(\pi) = \min\{\sigma_s / \min(V_s, \lambda(\pi - x)),\ \underbrace{(1 - \rho_a^0)\,\sigma_s/(\sigma_s + m)}_{(MFN)}\}$ and $\lambda \in (0, 1]$, as derived on the previous slide.

MFN clause, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds an MFN clause (ρ_a^0, m). Seed holds a capped SAFE with discount.

- The seed investor finances the shock if and only if $\ell - \sigma_a + m \le (1 - \rho_a^0) W(I)$.
- The clause binds, $\rho_s^* = (1 - \rho_a^0)\sigma_s^* / (\sigma_s^* + m)$, with the cap $V_s^* = (\sigma_s^* + m)/(1 - \rho_a^0)$, and the discount plays no role.
- The seed investor invests beyond its need, $\sigma_s^* > \ell - \sigma_a$ and $x^* < I$, if

$$(1 - \rho_a^0)\Big[\frac{m\, W(I)}{(\ell - \sigma_a + m)^2} + \frac{(\ell - \sigma_a)\big(1 - F(I)\big)}{\ell - \sigma_a + m}\Big] > 1,$$

  which holds on the smallest rounds.

MFN clause, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds an MFN clause (ρ_a^0, m). Seed holds a capped SAFE with discount.

The accelerator solves

$$\max_{\sigma_a,\, \rho_a^0,\, m \le \sigma_a} \Pi_a = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell + \int_{\sigma_a}^{\bar{\ell}} \left[W(I) - (\ell - \sigma_a) - \Pi_s^*(\ell) - \int_{x^*(\ell)}^{I} F(t)\, dt \right] g(\ell)\, d\ell - \sigma_a,$$

with $\bar{\ell}$ the largest shock the seed investor finances, $\bar{\ell} = \sigma_a + (1 - \rho_a^0) W(I) - m$.

MFN clause, capped SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

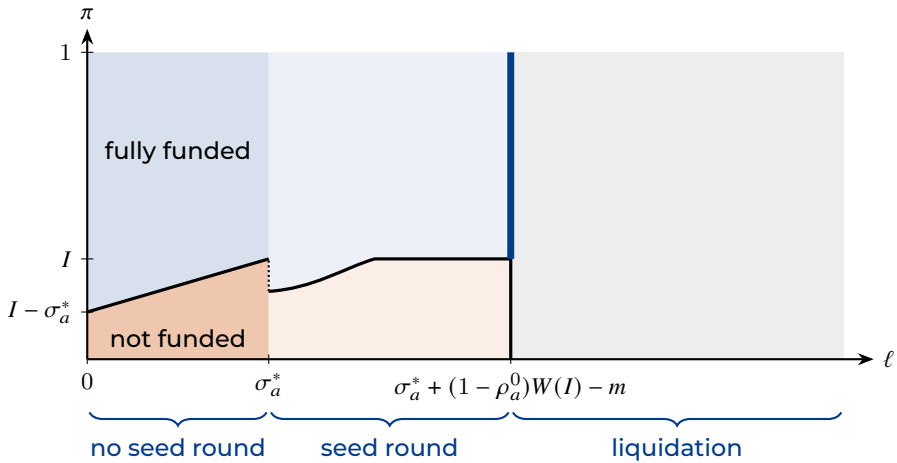
Accelerator holds an MFN clause (ρ_a^0, m). Seed holds a capped SAFE with discount.

If the optimal fixed share comes with a contribution, every optimum has $m^* > 0$. At given (ρ_a^0, m) with $m < \sigma_a^*$, the contribution solves

$$\left(\rho_a^0 W(I) + m\right) g(\bar{\ell}) = \left(1 - G(\bar{\ell})\right) + \int_0^{\sigma_a^*} F(I + \ell - \sigma_a^*)\, g(\ell)\, d\ell + \int_{\sigma_a^*}^{\bar{\ell}} \left(\Pi_s^*(\ell) + \int_{x^*(\ell)}^{I} F(t)\, dt\right) g'(\ell)\, d\ell,$$

with $\bar{\ell} = \sigma_a^* + (1 - \rho_a^0) W(I) - m$ the last shock the seed investor finances, on which the accelerator keeps $\rho_a^0 W(I) + m$.

MFN clause, capped SAFE



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MFN clause with the accelerator's SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.

Accelerator holds an MFN clause whose fixed part is its own SAFE, on $\sigma_a - m$. Seed holds a capped SAFE.

The fixed part of the clause can be a share or the accelerator's own SAFE: numerically, it makes no difference.

- The seed investor and the MFN component share what the accelerator's SAFE leaves, and the seed investor still prefers the capped SAFE.
- The fixed share is the limit of a SAFE of cap $(\sigma_a - m)/\rho_a^0$ as $m \to \sigma_a$.
- The value $(\sigma_a - m)/\lambda_a$ kept in the low states changes the seed investor's answer only on the small rounds where it invests beyond its need.
- In all configurations, it raises the accelerator's profit by at most $2 \cdot 10^{-5}$, the numerical precision.

Back

A project is fully funded above its threshold

The probability of full funding is

$$\int_0^{\sigma_a} \big(1 - F(I + \ell - \sigma_a)\big)\, g(\ell)\, d\ell + \int_{\sigma_a}^{\sigma_a + (1-\rho_a^0)W(I) - m} \big(1 - F(x^*)\big)\, g(\ell)\, d\ell.$$

- The seed investor finances $\sigma_a < \ell \leq \sigma_a + (1 - \rho_a^0)W(I) - m$, where the accelerator keeps $\rho_a^0 W(I) + m$ on the last project. A fixed share is $m = 0$, no clause is $\rho_a^0 = m = 0$.
- On a project the accelerator finances alone, the unused cash lowers the threshold below I.
- The seed investor covers only its need, $x^* = I$, except on small rounds under the MFN clause, $x^* < I$.
- A dollar more of contribution lowers the thresholds and moves the seed investor's zone up: full funding rises.

Back

The fixed share moves full funding either way

Moving from no clause to the optimal fixed share changes the probability of full funding by

$$\underbrace{\int_{\sigma_a^B}^{\sigma_a^P} \left[\int_0^s f(I+\ell-s)\, g(\ell)\, d\ell + \big(1-F(I)\big) g\big(s+W(I)\big) \right] ds}_{\text{contribution}} - \underbrace{\big(1-F(I)\big)\Big[G\big(\sigma_a^P + W(I)\big) - G\big(\sigma_a^P + (1-\rho_a^P)W(I)\big)\Big]}_{\text{last project} > 0}.$$

- The contribution term has the sign of $\sigma_a^P - \sigma_a^B$.
- If the fixed share does not raise the contribution, full funding falls.

Back

The MFN clause moves full funding either way

With a capped SAFE, moving from the fixed share to the MFN clause changes the probability of full funding by

$$\underbrace{\int_{\sigma_a^P}^{\sigma_a^M}\Big[\int_0^s f(I+\ell-s)\,g(\ell)\,d\ell+\big(1-F(I)\big)g\big(s+(1-\rho_a^P)W(I)\big)\Big]ds}_{\text{contribution}}$$

$$+\underbrace{\big(1-F(I)\big)\Big[G(\bar{\ell}^M)-G\big(\sigma_a^M+(1-\rho_a^P)W(I)\big)\Big]}_{\text{last project}}+\underbrace{\int_{\sigma_a^M}^{\bar{\ell}^M}\big(F(I)-F(x^*)\big)g(\ell)\,d\ell}_{\text{beyond the need}\,\geq\,0},$$

- $\bar{\ell}^M=\sigma_a^M+(1-\rho_a^{0M})W(I)-m^M$ is the last shock the seed investor finances under the MFN clause.
- The contribution term has the sign of $\sigma_a^M-\sigma_a^P$, the last-project term the sign of $\rho_a^P W(I)-\rho_a^{0M}W(I)-m^M$.
- Only the investment beyond the need is signed.

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A seed round covers at most $W(I) < I$

Accelerator	Probability of a seed round
No clause	$G\left(\sigma_a^B + W(I)\right) - G(\sigma_a^B)$
Fixed share	$G\left(\sigma_a^P + (1-\rho_a^P)W(I)\right) - G(\sigma_a^P)$
MFN clause, capped SAFE	$G\left(\sigma_a^M + (1-\rho_a^{0M})W(I) - m^M\right) - G(\sigma_a^M)$
MFN clause, priced SAFE	below the capped SAFE

- Whatever the protection, the seed investor finances a shock only if $\ell - \sigma_a \leq W(I) < I$.
- A protection keeps $\rho_a^0 W(I) + m$ for the accelerator on the last project, and shortens the zone by as much.
- A different contribution slides the zone: the probability rises or falls with the slope of g.

Back

The MFN clause moves seed rounds either way

With a capped SAFE, moving from the fixed share to the MFN clause changes the probability of a seed round by

$$\underbrace{\left[G\left(\sigma_a^M + (1-\rho_a^P)W(I)\right) - G(\sigma_a^M)\right] - \left[G\left(\sigma_a^P + (1-\rho_a^P)W(I)\right) - G(\sigma_a^P)\right]}_{\text{contribution}}$$

$$+ \underbrace{G(\bar{\ell}^M) - G\left(\sigma_a^M + (1-\rho_a^P)W(I)\right)}_{\text{last project}}.$$

- The last-project term has the sign of $\rho_a^P W(I) - \rho_a^{0M} W(I) - m^M$.
- The contribution term has no sign.

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The fixed share moves total surplus either way

Moving from no clause to the optimal fixed share changes total surplus by

$$\underbrace{\int_{\sigma_a^P}^{\sigma_a^B} \Big[\big(1 - G(s + W(I))\big) + \int_0^s F(I + \ell - s)\, g(\ell)\, d\ell\Big] ds}_{\text{contribution}} - \underbrace{\int_{\sigma_a^P + (1-\rho_a^P)W(I)}^{\sigma_a^P + W(I)} \big(\sigma_a^P + W(I) - \ell\big) g(\ell)\, d\ell}_{\text{last project} > 0}.$$

- Without clause, total surplus falls with the contribution.
- The contribution term has the sign of $\sigma_a^B - \sigma_a^P$.
- The fixed share stops the seed investor $\rho_a^P W(I)$ before $\sigma_a^P + W(I)$.

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The MFN clause moves total surplus either way

With a capped SAFE, moving from the fixed share to the MFN clause changes total surplus by

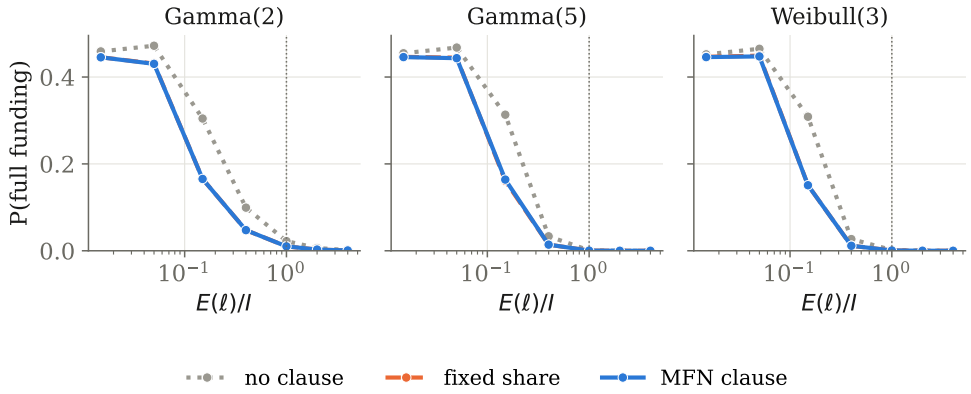
$$\underbrace{\int_{\sigma_a^P}^{\sigma_a^M} \left[\rho_a^P W(I)\, g\big(s+(1-\rho_a^P)W(I)\big) - \big(1-G(s+(1-\rho_a^P)W(I))\big) - \int_0^s F\, g\, d\ell\right] ds}_{\text{contribution}}$$

$$+ \underbrace{\int_{\sigma_a^M+(1-\rho_a^P)W(I)}^{\bar{\ell}^M} \big(\sigma_a^M + W(I) - \ell\big) g(\ell)\, d\ell}_{\text{last project}} - \underbrace{\int_{\sigma_a^M}^{\bar{\ell}^M} \int_{x^*}^{I} F\, g(\ell)\, d\ell}_{\text{beyond the need} \geq 0}.$$

- The last-project term has the sign of $\rho_a^P W(I) - \rho_a^{0M} W(I) - m^M$, the contribution term no sign.
- The investment beyond the need raises full funding, but lowers total surplus.

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Which projects survive: full funding



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